

Al-Usra Hajj and Umrah Company Management System

Database Project Report

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Introduction

The Al-Usra Hajj and Umrah Company Management System is a desktop database application designed to manage the main operations of a Hajj and Umrah company. The system was developed using Java Swing for the graphical user interface and MySQL as the relational database management system.

The main idea of the project is to organize the company's data in one centralized database instead of using manual files or separate records. The system manages users, employees, Hajj applicants, campaigns, hotels, transportation, bookings, employee tasks, and messages.

The application uses a role-based login system. This means that each user enters the system using a username and password, and the system opens a different dashboard depending on the user role. The three main roles are Admin, Employee, and Customer.

The project demonstrates important database concepts such as tables, primary keys, foreign keys, relationships, constraints, SQL queries, CRUD operations, and Java database connectivity using JDBC.

Problem Statement

Hajj and Umrah companies deal with many types of data, such as customer information, employee records, Hajj applications, hotel details, transportation information, packages, bookings, tasks, and internal messages. Managing this data manually can cause many problems, including data duplication, missing records, slow searching, and difficulty in updating information.

Another important problem is access control. In a real company, not all users should have the same permissions. The Admin should manage company data, Employees should manage only their tasks and messages, and Customers should access only booking-related services.

For this reason, the system was created to provide a centralized database and a user-friendly graphical interface. The system helps organize data, reduce manual work, improve accuracy, and provide role-based access for different users.

Project Objectives

The main objectives of this project are:

1. To design a database system for managing a Hajj and Umrah company.
2. To store company data in a structured MySQL database.
3. To create a Java Swing graphical user interface.
4. To provide a role-based login system for Admin, Employee, and Customer.
5. To allow the Admin to manage employees, campaigns, Hajj applicants, hotels, and transportation companies.
6. To allow Employees to view their profiles, manage assigned tasks, and read messages.

7. To allow Customers to access booking-related services.
8. To apply CRUD operations: Create, Read, Update, and Delete.
9. To connect Java with MySQL using JDBC.
10. To make the system easier to use and more organized.

System Users and Roles

The system is based on three main user roles: Admin, Employee, and Customer. Each role has a different dashboard and different permissions.

Admin

The Admin is responsible for managing the main company data. The Admin can manage campaigns, Hajj applicants, employees, hotels, and transport companies. This role has the highest level of access in the system.

Employee

The Employee can access personal work-related features. The employee can view and update profile information, view assigned tasks, mark tasks as done, and read messages. The employee does not have full access to company data like the Admin.

Customer

The Customer can access services related to bookings, hotels, and transportation. The customer has limited access and cannot manage internal company data.

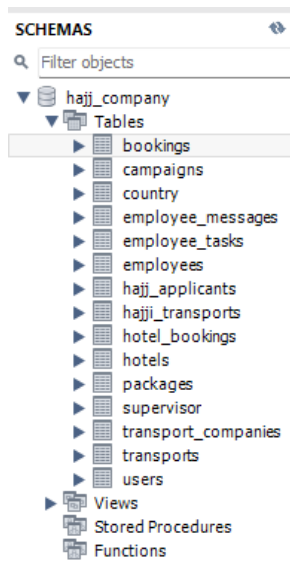
Main Functions	Role
Manage campaigns, applicants, employees, hotels, and transport companies	Admin
View profile, manage tasks, and read messages	Employee
Access booking, hotel, and transportation services	Customer

Database Design

The system uses a MySQL database named hajj_company. The database contains several tables, and each table represents a specific entity in the system. The database was designed as a relational database because the system contains different types of data that

are connected to each other.

The main tables in the database are:



This design keeps the data organized and reduces duplication. For example, employee information is stored in the employees table, while employee tasks are stored separately in the employee_tasks table and connected using emp_id.

Relationships and Constraints

The database uses relationships to connect tables together. These relationships are created using primary keys and foreign keys.

A primary key uniquely identifies each record in a table. For example, emp_id uniquely identifies each employee in the employees table, and task_id uniquely identifies each task in the employee_tasks table.

A foreign key connects one table to another. For example, the emp_id column in the employee_tasks table refers to the emp_id column in the employees table. This means that each task is connected to a specific employee.

Examples of relationships in the system include:

Type	Relationship
One-to-Many	One employee has many tasks
One-to-Many	One employee has many messages
One-to-Many	One hotel has many hotel bookings
One-to-Many	One transport company has many transports
Many-to-Many	Hajj applicants and transports are connected through <code>hajji_transports</code>

The system also uses constraints to keep data valid. Examples include:

- **PRIMARY KEY:** prevents duplicate record identifiers.
- **FOREIGN KEY:** maintains relationships between tables.
- **NOT NULL:** prevents empty required fields.

- UNIQUE: prevents duplicate usernames.
- CHECK: restricts values, such as allowing only admin, employee, or customer as user roles.
- AUTO_INCREMENT: automatically generates numeric IDs.

The database also uses actions such as ON DELETE CASCADE and ON DELETE SET NULL. ON DELETE CASCADE deletes related child records when the parent record is deleted. ON DELETE SET NULL keeps the child record but sets the foreign key value to NULL.

SQL Implementation

The SQL script is responsible for creating the database, creating the tables, adding constraints, inserting sample data, and creating relationships between tables.

The script begins by creating the database:

```
3 • CREATE DATABASE IF NOT EXISTS `hajj_company`
4 • USE `hajj_company`;
```

The CREATE DATABASE command creates the database if it does not already exist. The USE command selects the database so that the following SQL commands are applied inside it.

The script also creates tables using CREATE TABLE. For example, the users table stores login information:

```
CREATE TABLE `users` (
  `id` INT AUTO_INCREMENT NOT NULL,
  `username` VARCHAR(50) NOT NULL,
  `password` VARCHAR(100) NOT NULL,
  `role` VARCHAR(20) NOT NULL,
  CONSTRAINT `users_role_check` CHECK (`role` IN ('admin', 'employee', 'customer')),
  CONSTRAINT `users_pkey` PRIMARY KEY (`id`),
  CONSTRAINT `users_username_key` UNIQUE (`username`)
)
```

This table contains the user ID, username, password, and role. The id is auto-incremented, the username is unique, and the role is restricted to three possible values.

The project uses SQL operations such as:

INSERT

```
• INSERT INTO `users` (`id`, `username`, `password`, `role`) VALUES (1, 'admin', 'admin123', 'admin');
```

SELECT , UPDATE , DELETE .

Java Implementation and JDBC

The system was developed using Java Swing in NetBeans. Java Swing was used to create the graphical user interface, including forms, buttons, tables, labels, and panels.

The connection between Java and MySQL is done using JDBC. JDBC allows the Java application to connect to the database and execute SQL queries.

The main database connection is handled in the DBConnection.java class. The connection URL is similar to:

```
jdbc:mysql://localhost:3306/hajj_company
```

This URL means that Java connects to a MySQL database running on the local computer using port 3306, and the database name is hajj_company.

The system uses:

- Connection: to connect to the database.
- PreparedStatement: to execute SQL queries safely.
- ResultSet: to store results returned from SELECT queries.
- JTable: to display database records inside the GUI.

The general flow of the system is:

Button Click → SQL Query → MySQL Database → ResultSet → GUI Display

For example, when the employee opens the My Tasks screen, the program sends a SELECT query to the database, retrieves the tasks assigned to the logged-in employee, and displays them inside a JTable.

The project also uses PreparedStatement instead of directly writing user input inside SQL queries. This makes the system safer and reduces the risk of SQL injection.

Main System Screens

The system contains several graphical user interfaces. Each screen is designed for a specific purpose.

Login Form

The Login Form is the first screen in the system. The user enters a username and password. The system checks the information from the users table and opens the correct dashboard depending on the user role.

Admin Dashboard

The Admin Dashboard allows the Admin to access management screens such as campaigns, applicants, employees, hotels, and transport companies.

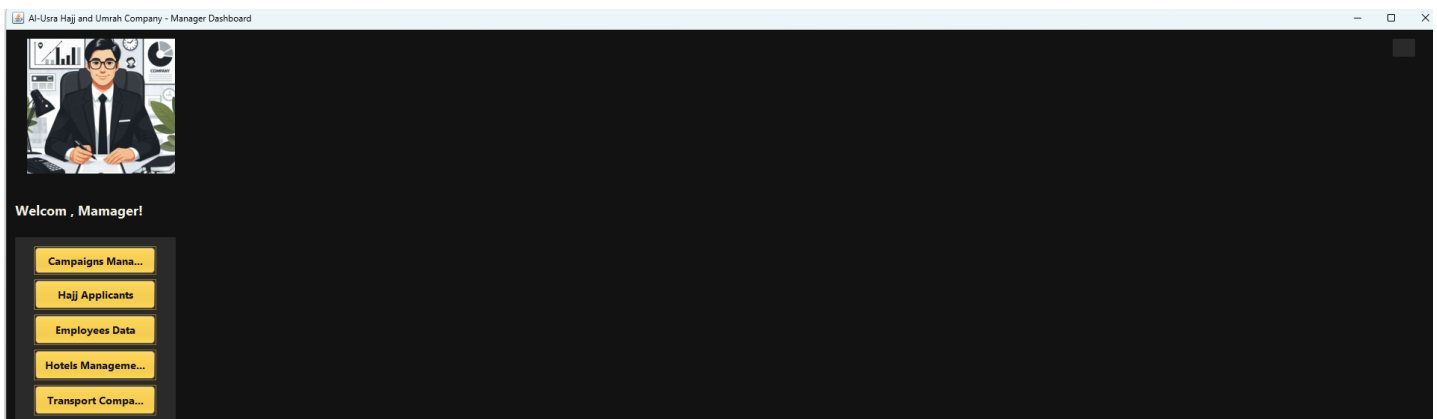
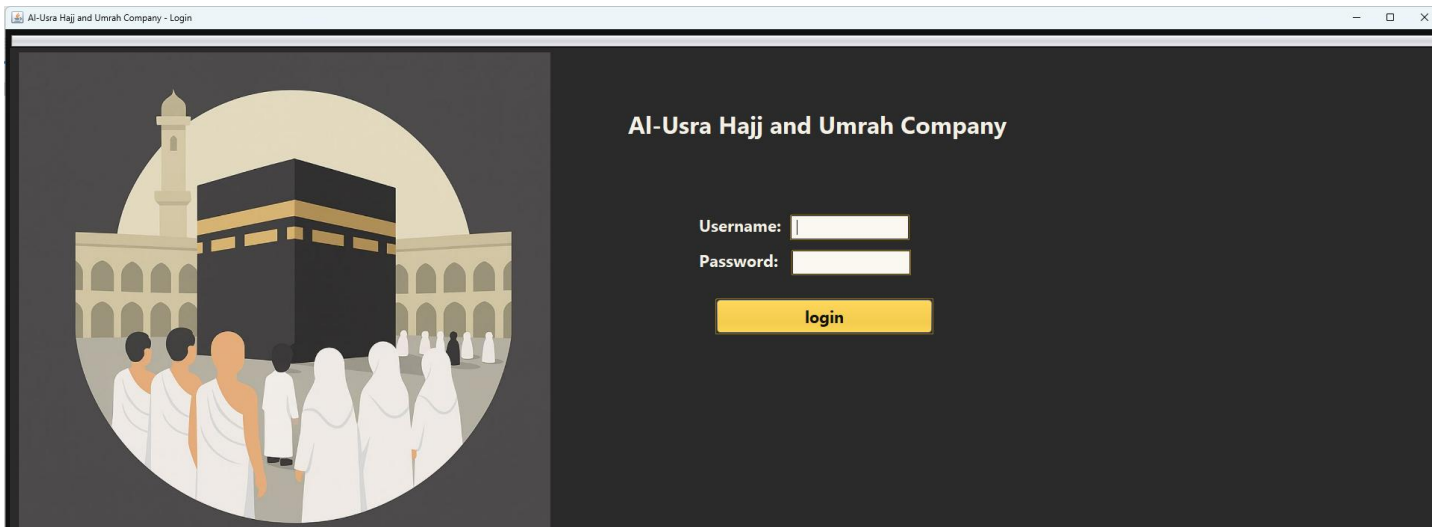
Employee Dashboard

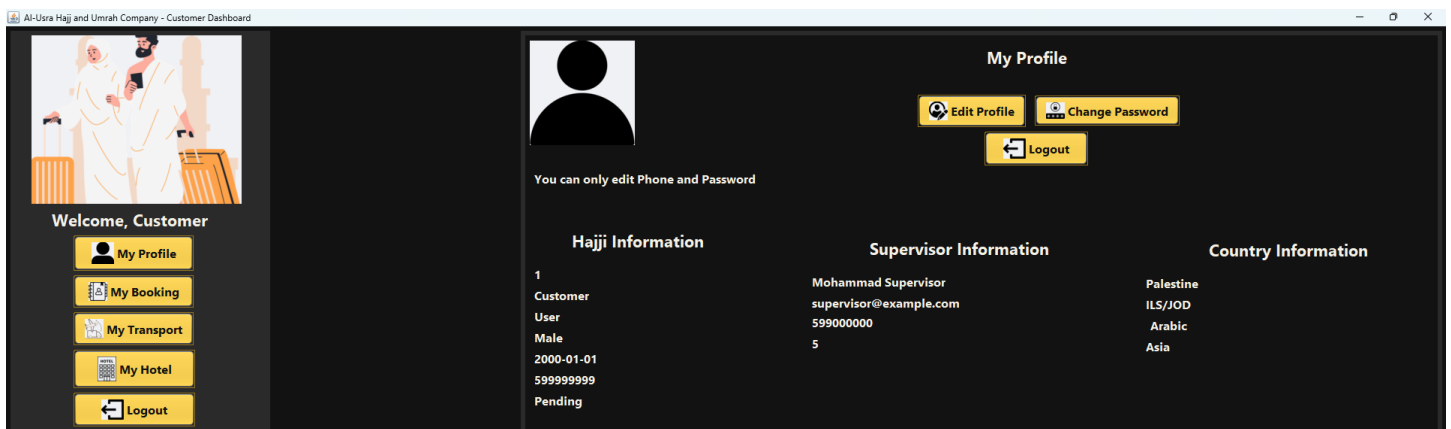
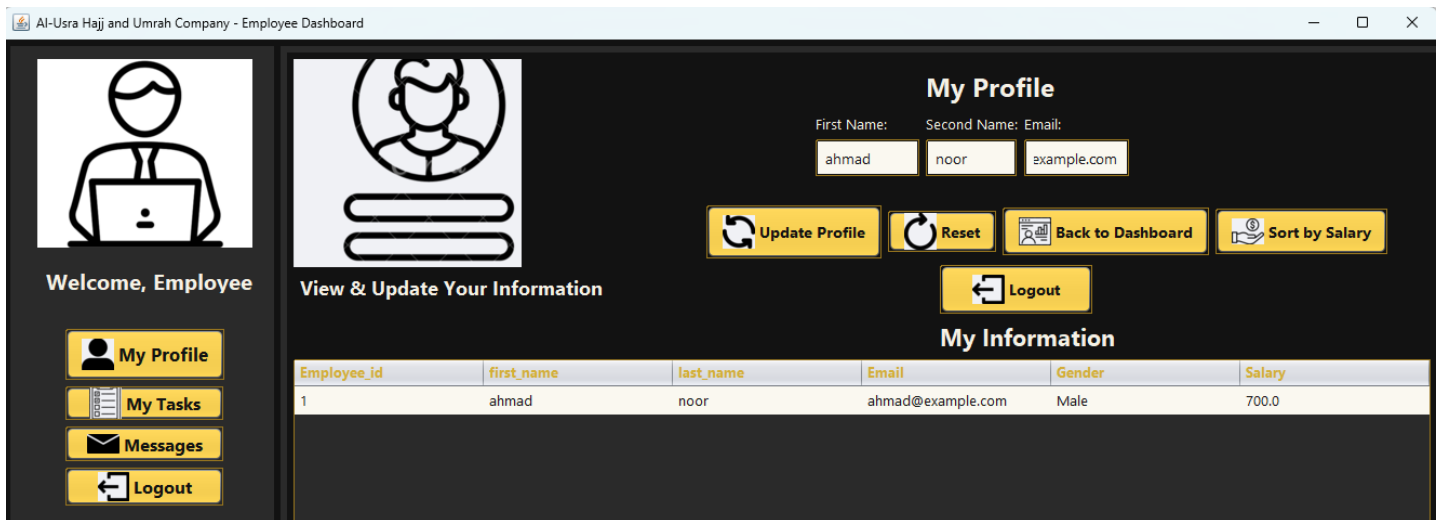
The Employee Dashboard allows employees to access their profile, tasks, and messages. The My Tasks screen is important because it shows only the tasks assigned to the logged-in employee.

Customer Dashboard

The Customer Dashboard allows customers to access booking-related services such as packages, hotel services, and transportation services.

The GUI was improved to be clearer and more organized. A dark and gold theme was used to make the design more professional and suitable for a Hajj and Umrah company system.





Conclusion

In conclusion, the Al-Usra Hajj and Umrah Company Management System successfully provides a database-based solution for managing the main operations of a Hajj and Umrah company. The system organizes data in a MySQL relational database and provides a Java Swing graphical interface for different users.

The project supports role-based access, allowing Admins, Employees, and Customers to use different dashboards according to their responsibilities. It also applies important database concepts such as primary keys, foreign keys, relationships, constraints, CRUD operations, and JDBC connectivity.

Overall, the project demonstrates how a real-world company system can be designed and implemented using database principles and Java programming.